

R80 Wideband/Multi-mode PLL Receiver

User Manual V1.0

Brief Introduction

The R80 is an original wideband/multi-mode PLL receiver from the Taobao store “Dinuco Radio.” This unit uses a classic analog superheterodyne architecture and is divided into two bands. It uses single conversion for 2 – 29.999 MHz and double conversion for 30 – 1000 MHz. FM and AM reception in this unit is completed by the TA8132, narrowband FM is completed by the MC3371, and SSB/CW reception is completed by the TA7765 and MC3371. At the same time, to improve practicality, the unit has a built-in real-time clock and spectrum display function.

The appearance is as follows:



Main Specifications

Power supply: 7.5 V

Antenna: Band 0: 2 – 29.999 MHz, high-impedance input, unbalanced;

Band 1: 30 – 10000 MHz, 50 ohm, unbalanced
 Typical receiving current: 630 mA
 Receiving frequency: 2 – 1000 MHz
 Operating modes: FM, AM, nFM (narrowband FM), USB, LSB, CW
 Receiving sensitivity: FM/AM > -100 dBm; nFM > -110 dBm; SSB/CW > -120 dBm
 Step: Band 0: 2 – 29.999 MHz, 10/100/1000/1000 Hz;
 Band 1: 30 – 10000 MHz, 1K/10K/100K/1MHz
 Audio output: 8 ohm, 250 mW

Circuit Principle

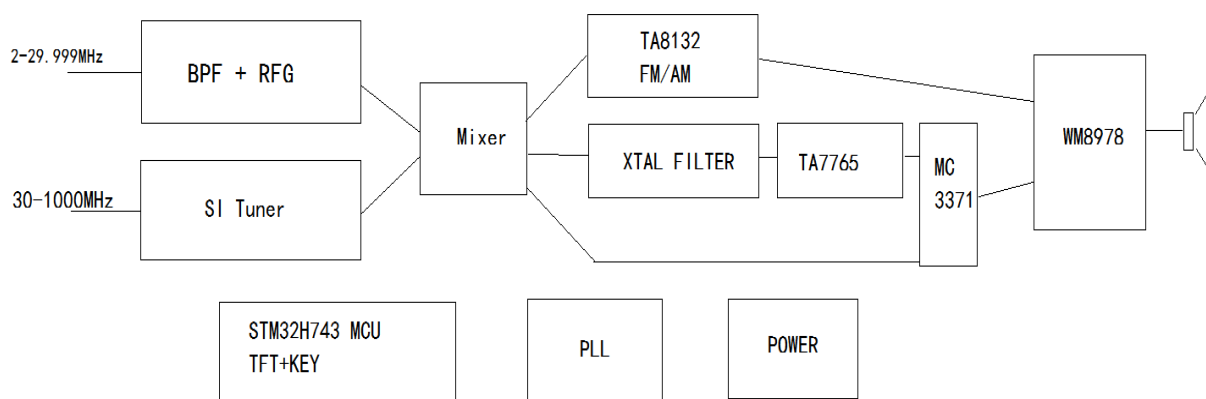
See the figure below. This unit has two antenna connectors. For signals below 30 MHz, the signal received by the antenna first passes through a band-pass filter and an RF amplifier (RFG), and is then sent into the MIXER for mixing.

Depending on the signal mode: if it is an FM signal, the mixer outputs 10.7 MHz to the TA8132 for FM stereo reception and AM reception; if it is an nFM signal, it outputs a 455 kHz IF to the MC3371; if it is an SSB or CW signal, it outputs a 10.7 MHz signal to the IF amplifier TA7765, and then to the MC3371 for product detection. All audio signals are finally collected and amplified by the WM8978, and at the same time the audio signals are uploaded to the MCU via the I2S interface.

For 30 MHz signals, reception is handled uniformly by the SiTuner, which outputs a 36 MHz IF to the Mixer. The processing flow is the same as that of the low-frequency band.

The MCU of this unit uses the STM32H743, which handles TFT screen display, keyboard, and infrared processing. It also performs FFT on the uploaded audio signal and displays the spectrum on the TFT screen.

In addition, this unit also has an external TF card, and the recording function is implemented by the STM32H743.



Appearance Introduction

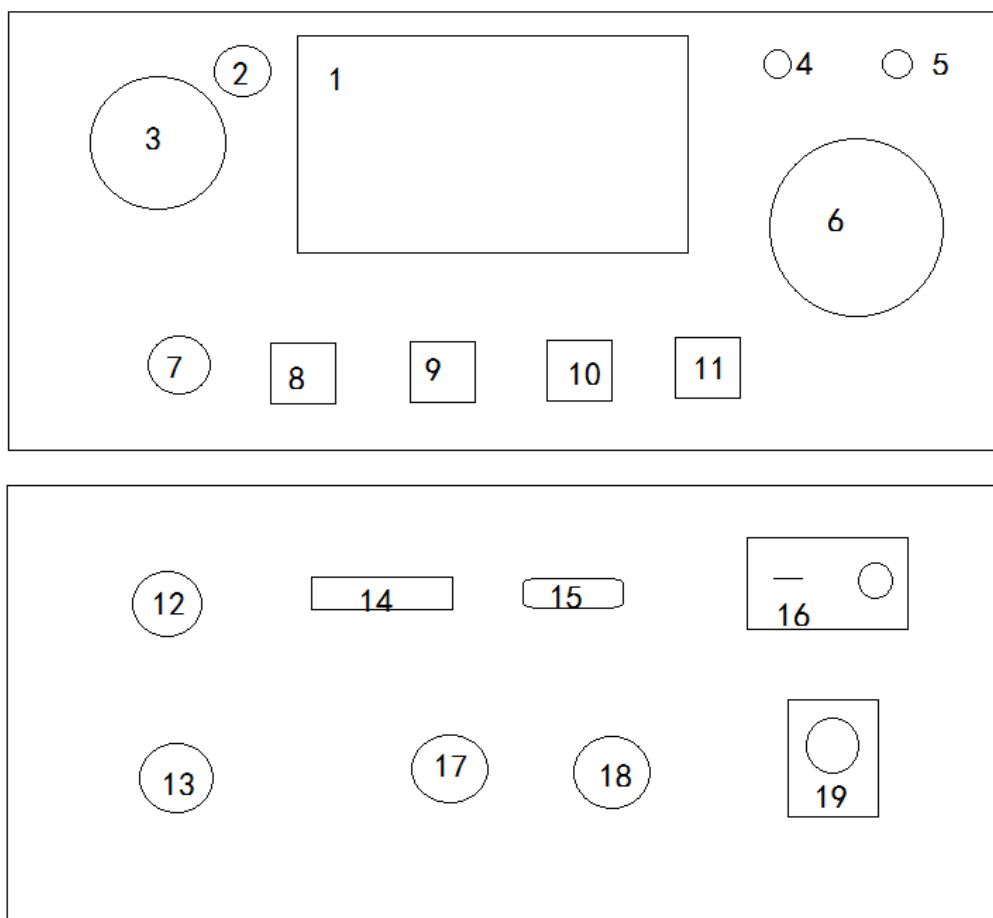
1 Front panel



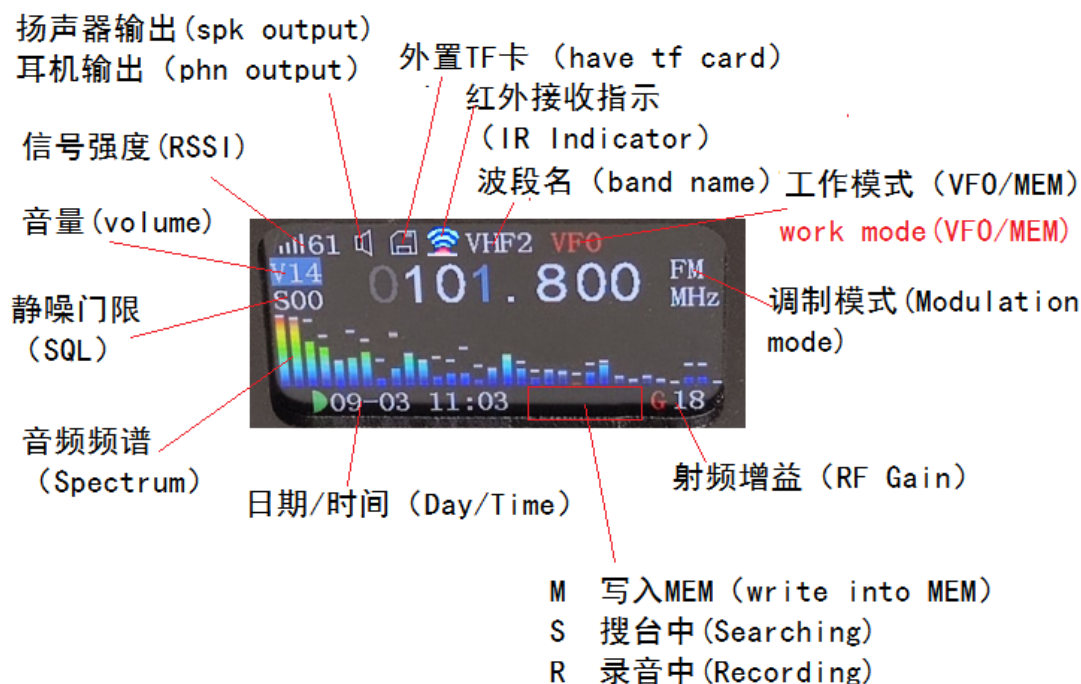
2 Rear panel



3 Rear Panel Function Description



1 TFT screen



2 IR receiver

3 Volume/squelch adjustment knob. Short press to switch between volume and squelch; long press (>3 seconds) to switch audio output between the internal speaker and headphones.

4 Stereo signal indicator

5 Receive indicator; it lights up when the received signal exceeds the squelch threshold.

6 Frequency and RF gain adjustment knob. Short press to switch between frequency and RF gain.

7 Headphone output jack.

8 Short press: Menu key;

Long press: VFO/MEM mode switch. VFO mode is the default after power-on.

9 Short press: Step key;

Long press: Store the current frequency in MEM; a new entry will be added to the CH table.

10 Short press: Band switch, switching between <30 MHz and ≥30 MHz. Note that different bands use different antennas.

Long press: Start/stop auto search in the current band.

11 Short press: Mode switch, switching among AM, FM, nFM, USB, LSB, and CW.

Long press: Start/stop recording.

12 Antenna connector for Band 0 <30 MHz signal input

13 Antenna connector for Band ≥30 MHz signal input

14 External TF card. A card is already inserted by default.

15 USB interface (Type-C), used to connect to a computer. Currently it is only for firmware upgrade.

16 Power switch.

17/18 Audio signal line output; the level cannot be adjusted.

19 Power socket, 7.5 V.

Note: Volume, SQL threshold, and RFG settings are all between 0 and 25.

Operating Instructions

After normal power-on, the TFT screen displays normally, and the encoder is in frequency adjustment mode by default. At this time, press the STEP key, and you will see the light blue digits keep moving, indicating that the step is changing continuously.

The small knob is in volume adjustment mode by default after power-on. Press the VOL/SQL knob to switch back and forth between VOL and SQL settings. In VOL setting, the speaker icon is as shown below; in SQL setting, the letter S is as shown below. They respectively indicate the current encoder knob mode.

VOL set mode 

SQL set mode 

Press the MENU key, and the display will switch to the CH menu.  indicates the currently selected item; turning the encoder will change the selection.



In the CH menu, there are two CH entries by default. The four keys are, in order:

MENU: Press once to switch to the next menu.

DELT: Delete key. If the selected item is not locked, pressing this key will delete the current CH entry. The CH table must have at least one entry and cannot be completely emptied.

LOCK: Lock key. Pressing this key locks the currently selected CH entry, and it cannot be deleted. Press it again to unlock.

Press the MENU key, and the display will switch to the RECORD menu.  indicates the

currently selected item; turning the encoder will change the selection.



In the RECORD menu, the four keys are as follows:

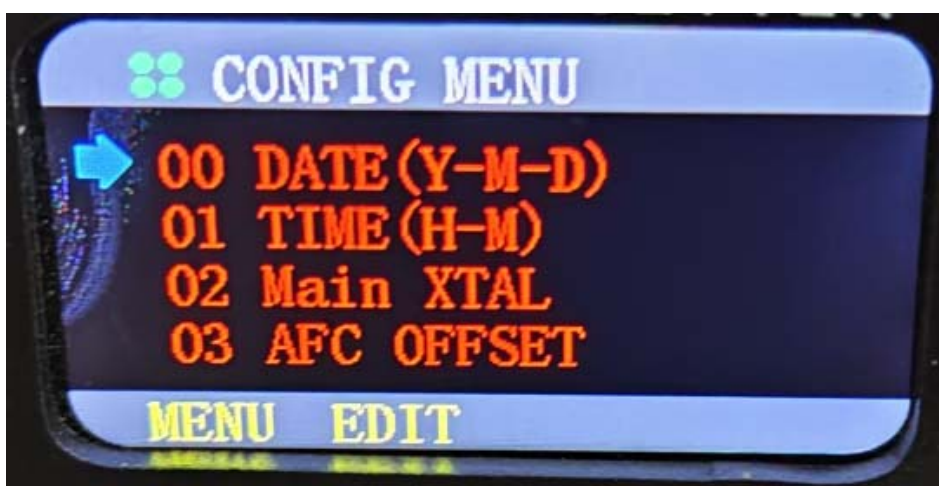
MENU Press once to switch to the next menu.

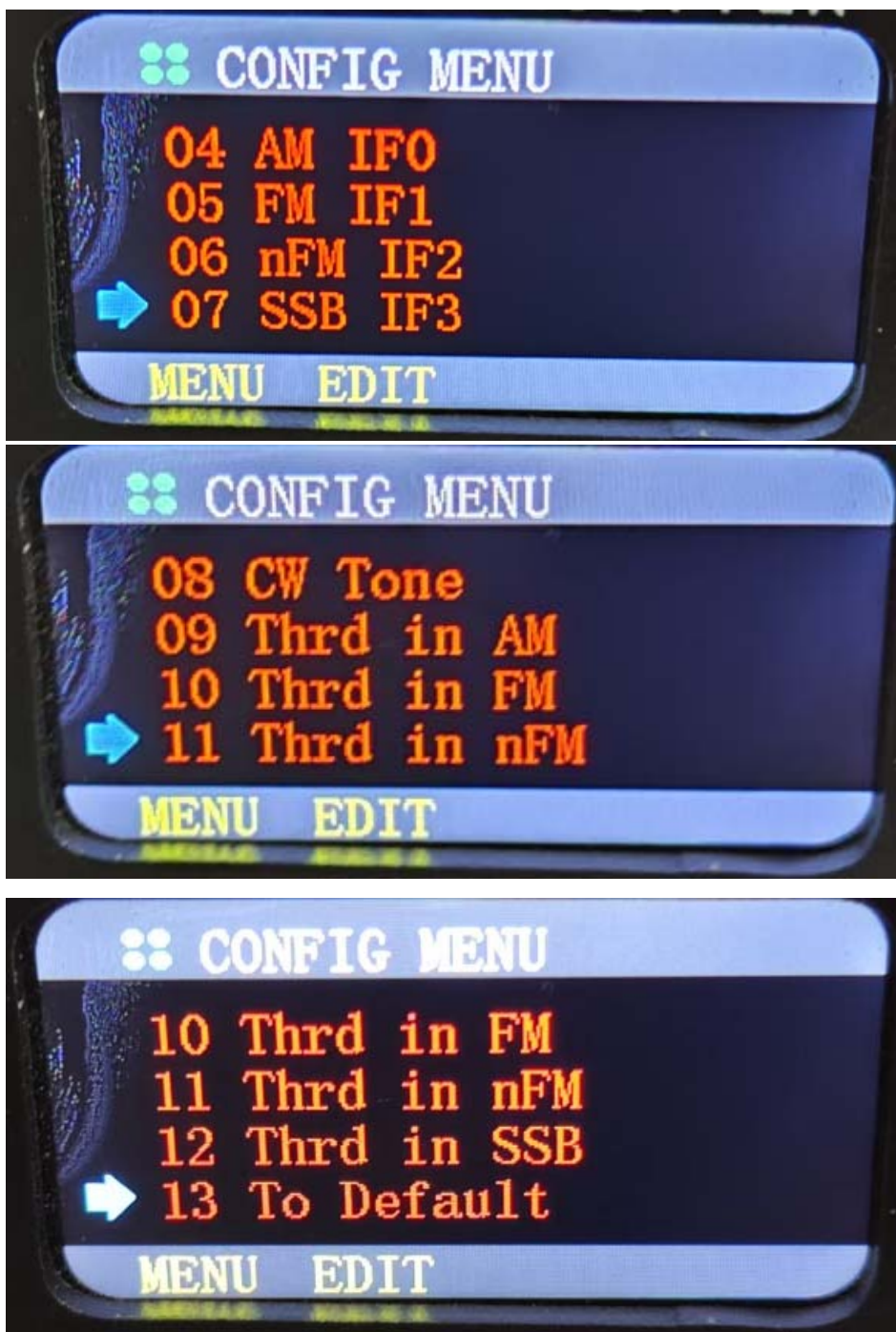
DELT Delete key. Pressing this key will delete the currently selected RECORD.

PLAY Play key. Pressing this key will play the currently selected WAV file. And you can also stop the play process by this key.

Note: If you need to copy WAV files from the unit's TF card, remove the TF card and use a card reader to read the internal WAV files.

Press the MENU key to enter the configuration menu.  indicates the currently selected item; turning the encoder will change the selection. There are currently items 0–13.





There are 2 keys in this menu. MENU: Press once to switch to the next menu. EDIT: Press once to edit the selected item (turn the encoder to modify), and press again to finish editing. This screen is intended for experienced personnel with instruments.

0 Date, in year-month-day format. Press EDIT to enter edit mode. At this time, TAB will be displayed.

After selecting the encoder, input the year, then press the TAB key to edit the month. After inputting the date, press EDIT to exit the modification.



1 Time, in hour-minute format, 24-hour system. Press EDIT to edit, and press TAB to switch.



2 Main XTAL frequency value, default 16000000, unit Hz. Press EDIT to edit.



3 AFC offset parameter. Because the SiTuner's internal PLL step has an error depending on the temperature at the operating location, when the unit works in 30–1000 MHz, SSB/CW mode, it may be necessary to calibrate the following parameter. Other bands and other modes do not require it.

The following display is -4800, unit Hz, which means the main operating frequency will auto add -4800 Hz to the set frequency.



4 AM IF frequency value, default 10700000, unit Hz.



5 FM IF frequency value, default 10700000, unit Hz.

6 nFM IF frequency value, default 455000, unit Hz.

7 SSB IF frequency value, default 10698700, unit Hz. This is the center frequency of the internal crystal filter.

8 CW sidetone frequency, default 800, unit Hz.



9 Use AM mode, the search signal strength threshold value, default 230, means that automatic search is valid when the AD sample exceeds 230. If all stations are weak, this parameter can be appropriately lowered.



10 Use FM mode, the search signal strength threshold value.

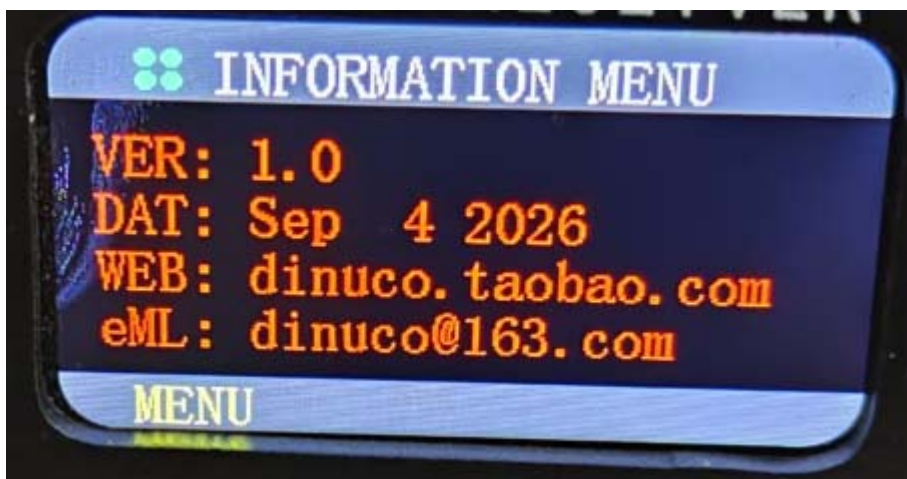
11 Use nFM mode, the search signal strength threshold value.

12 Use SSB mode, the search signal strength threshold value.

13 Restore factory settings. Press EDIT and the unit will restore factory settings.



Press the MENU key to enter the information menu.



Press MENU again to exit to the initial menu.

Remote Control Instructions

The remote control interface provided with this unit is as follows, and it can only be used in the initial menu.

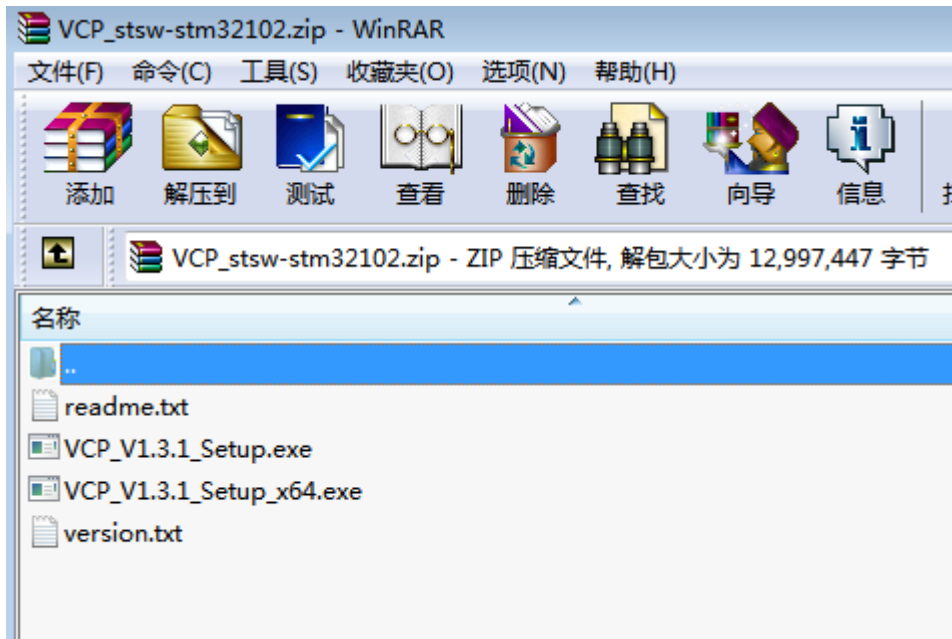


Firmware Upgrade

This unit supports self-upgrade of firmware. After DINUCO releases a new firmware version, customers can upgrade the firmware by themselves. The upgrade method is as follows.

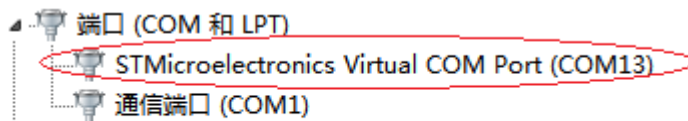
Note: If you do not have the correct firmware file, do not perform the upgrade steps!

1: In Win7 and earlier environments, install ST's virtual COM port driver. It is not required in Win10 and above.



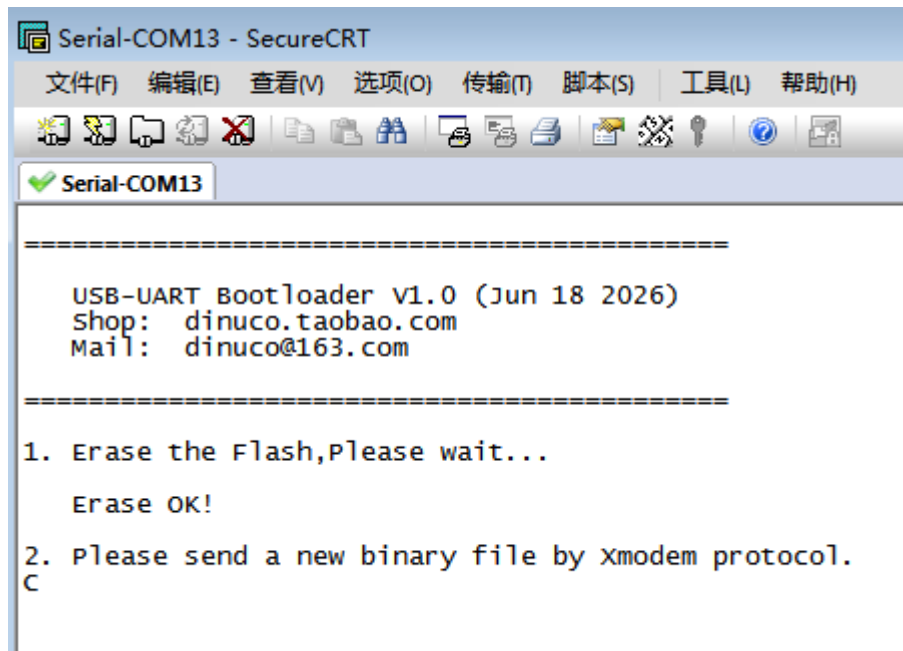
2 After installing the virtual driver, turn off the unit's power and connect the unit's USB interface to the computer's USB interface. At the same time, **hold down the unit's MENU and STEP keys**, then turn on the unit's power.

At this time, a virtual serial port created by the Loader program can be seen on the computer.

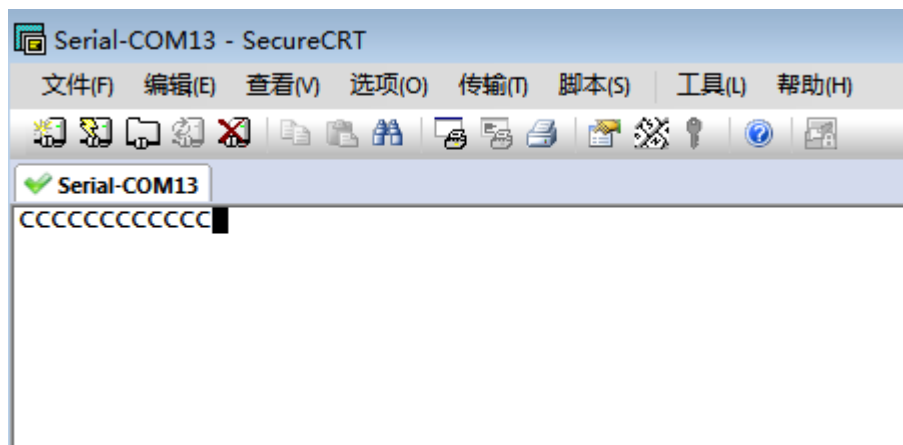


Note: This virtual serial port only appears after starting the Loader. There is no virtual serial port during normal use of the unit.

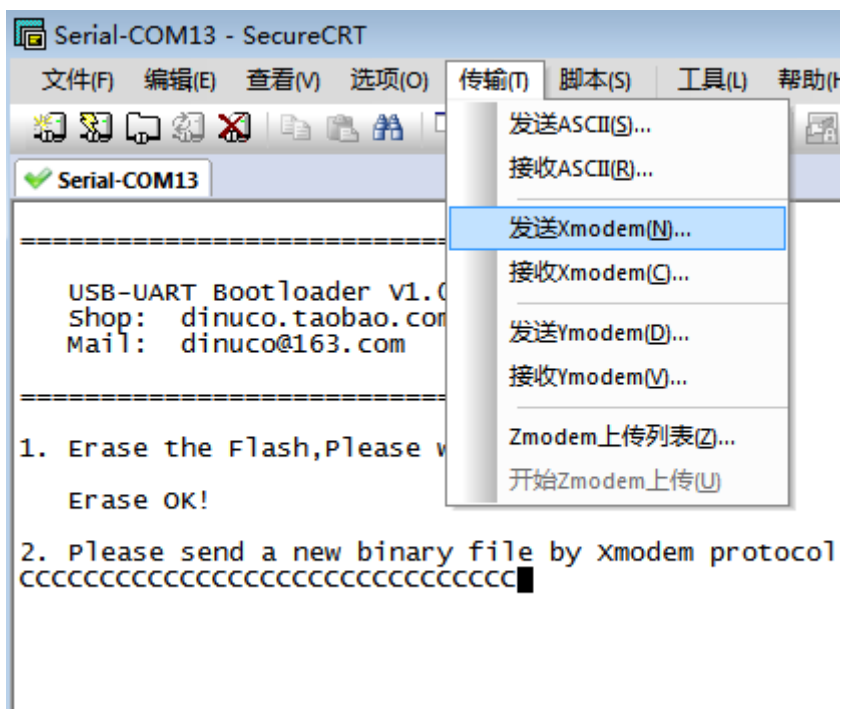
3 Use any terminal software, such as SecureCRT, and the following interface will appear:



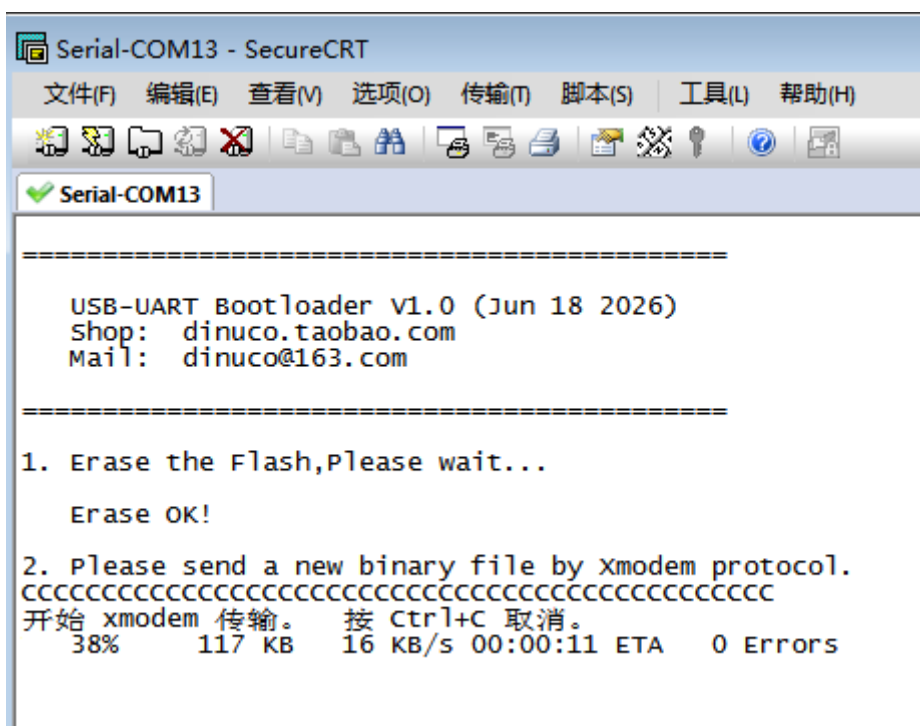
Or, because opening the serial port is slow, the following interface is displayed. As long as the character C appears, it means the firmware can be upgraded normally.



4 Select Send XMODEM and open the firmware file.



5 Start transferring the firmware file.



6 After the firmware file transfer is complete, the unit will automatically restart and enter the initial interface.